

NAME: Vaibhav Mangla

REG. NO.: 24BAI10690

VIT Bhopal University

1.

```
db.Employees.aggregate([
{
$lookup: {
from: "Departments",
localField: "dept_id",
foreignField: "dept_id",
as: "dept"
}
},
{ $unwind: "$dept" },
{
}
])
$project: {
_id: 0,
employee_name: 1,
department_name: "$dept.dept_name",
location: "$dept.location_id"
}
```

OUTPUT:

```
[
{ employee_name: "Alice Smith (CEO)", department_name: "Consulting", location: "NYC"
```

```

},
{ employee_name: "Bob Johnson", department_name: "Engineering", location: "SF" },
{ employee_name: "Charlie Brown", department_name: "Engineering", location: "SF" },
{ employee_name: "Diana Prince", department_name: "R&D", location: "London" },
{ employee_name: "Evan Clark", department_name: "R&D", location: "London" }
]
*/

```

2.

```

db.Projects.aggregate([
{
  $lookup: {
    from: "Departments",
    localField: "dept_id",
    foreignField: "dept_id",
    as: "dept"
  }
},
{ $unwind: "$dept" },
{
  $match: {
    "dept.dept_name": "Consulting"
  }
},
{
  $lookup: {
    from: "Assignments",

```

```

    localField: "project_id",
    foreignField: "project_id",
    as: "assignments"
  }
},
{
  $project: {
    _id: 0,
    project_name: 1,
    client_name: 1,
    total_hours: { $sum: "$assignments.hours_worked" }
  }
})

```

OUTPUT:

```

[
  {
  }
]

```

3.

```

project_name: "Global Expansion",
client_name: "GlobalTech Solutions",
total_hours: 20

db.Assignments.aggregate([
  {
    $lookup: {

```

```
from: "Projects",
localField: "project_id",
foreignField: "project_id",
as: "project"
}
},
{ $unwind: "$project" },
{
  $match: {
    "project.client_name": "GlobalTech Solutions"
  }
},
{
  $lookup: {
    from: "Employees",
    localField: "employee_id",
    foreignField: "employee_id",
    as: "employee"
  }
},
{ $unwind: "$employee" },
{
}
])
$project: {
  _id: 0,
```

```
employee_name: "$employee.employee_name",  
project_name: "$project.project_name"  
}
```

OUTPUT:

```
[  
  { employee_name: "Bob Johnson", project_name: "Aurora Migration" },  
  { employee_name: "Charlie Brown", project_name: "Aurora Migration" },  
  { employee_name: "Alice Smith (CEO)", project_name: "Global Expansion" },  
  { employee_name: "Fiona Glen", project_name: "Aurora Migration" }  
]
```

4.

```
db.Departments.aggregate([  
  {  
    $lookup: {  
      from: "Employees",  
      localField: "dept_id",  
      foreignField: "dept_id",  
      as: "employees"  
    }  
  },  
  {  
    $match: {  
      employees: { $size: 0 }  
    }  
  },  
  {
```

```
}  
])  
$project: {  
  _id: 0,  
  dept_name: 1  
}
```

OUTPUT:

```
[  
]
```

(No department without employees)

5.

```
db.Employees.aggregate([  
  {  
    $lookup: {  
      from: "Employees",  
      localField: "employee_id",  
      foreignField: "manager_id",  
      as: "reports"  
    }  
  },  
  {  
    $lookup: {  
      from: "Departments",  
      localField: "dept_id",  
      foreignField: "dept_id",  
      as: "dept"    }  
  }  
])
```

```

}
},
{ $unwind: "$dept" },
{
  $project: {
    _id: 0,
    manager_name: "$employee_name",
    department_name: "$dept.dept_name",
    direct_reports: { $size: "$reports" }
  }
}
])

```

OUTPUT:

```

[
  { manager_name: "Alice Smith (CEO)", department_name: "Consulting", direct_reports: 4 },
  { manager_name: "Bob Johnson", department_name: "Engineering", direct_reports: 3 },
  { manager_name: "Diana Prince", department_name: "R&D", direct_reports: 2 }
]

```

6.

```

db.Assignments.aggregate([
  {
    $group: {
      _id: "$employee_id",
      avg_hours: { $avg: "$hours_worked" }
    }
  }
]

```

```
},  
{  
  $lookup: {  
    from: "Employees",  
    localField: "_id",  
    foreignField: "employee_id",  
    as: "employee"  
  }  
},  
{ $unwind: "$employee" },  
{  
  $project: {  
    _id: 0,  
    employee_name: "$employee.employee_name",  
    avg_hours: 1  
  }  
}  
])
```

OUTPUT:

```
[  
  { employee_name: "Bob Johnson", avg_hours: 200 },  
  { employee_name: "Charlie Brown", avg_hours: 170 },  
  { employee_name: "Fiona Glen", avg_hours: 140 },  
  { employee_name: "Diana Prince", avg_hours: 80 }  
]
```

7.


```
var companyAvg = db.Employees.aggregate([
```

```
{
```

```
}
```

```
$group: {
```

```
  _id: null,
```

```
  avgSalary: { $avg: "$salary" }
```

```
}
```

```
]).toArray()[0].avgSalary;
```

```
db.Employees.aggregate([
```

```
{
```

```
$group: {
```

```
  _id: "$dept_id",
```

```
  deptAvg: { $avg: "$salary" }
```

```
}
```

```
},
```

```
{
```

```
$match: {
```

```
  deptAvg: { $gt: companyAvg }
```

```
}
```

```
},
```

```
{
```

```
$lookup: {
```

```
  from: "Departments",
```

```
  localField: "_id",
```

```
  foreignField: "dept_id",
```

```
  as: "dept"
```

```

}
},
{ $unwind: "$dept" },
{
}
])
$project: {
  _id: 0,
  department_name: "$dept.dept_name",
  deptAvg: 1
}

```

OUTPUT:

```

[
{ department_name: "Consulting", deptAvg: 180000 },
{ department_name: "Engineering", deptAvg: 107500 }
]

```

8.

```

db.Assignments.aggregate([
{
  $lookup: {
    from: "Projects",
    localField: "project_id",
    foreignField: "project_id",
    as: "project"
  }
},

```

```

{ $unwind: "$project" },
{
  $group: {
    _id: "$project.client_name",
    total_hours: { $sum: "$hours_worked" }
  }
},
{
}
])
$match: {
  total_hours: { $gt: 500 }
}

```

OUTPUT:

```

[
  { _id: "GlobalTech Solutions", total_hours: 470 },
  { _id: "Zenith Corp", total_hours: 855 }
]

```

9.

```

db.Employees.aggregate([
  {
    $group: {
      _id: "$dept_id",
      emp_count: { $sum: 1 }
    }
  },

```

```

{
  $lookup: {
    from: "Departments",
    localField: "_id",
    foreignField: "dept_id",
    as: "dept"
  },
  { $unwind: "$dept" },
  {
    $sort: {
      "dept.location_id": 1,
      emp_count: -1
    }
  },
  {
    $group: {
      _id: "$dept.location_id",
      department: { $first: "$dept.dept_name" },
      employee_count: { $first: "$emp_count" }
    }
  }
})

```

OUTPUT:

```

[
  { _id: "London", department: "R&D", employee_count: 4 },

```

```
{_id: "NYC", department: "Consulting", employee_count: 1 },  
{_id: "SF", department: "Engineering", employee_count: 4 }  
]
```

10.

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$project_id",  
      employees: { $addToSet: "$employee_id" }  
    }  
  },  
  {  
    $project: {  
      employee_count: { $size: "$employees" }  
    }  
  },  
  {  
    $match: {  
      employee_count: { $gt: 5 }  
    }  
  }  
])
```

OUTPUT:

```
[  
]
```

(No project has more than 5 employees)

11.

```
var rndProjects = db.Projects.find(
{ dept_id: 30 },
{ project_id: 1, _id: 0 }
).toArray().map(p => p.project_id);
db.Assignments.aggregate([
{
  $group: {
    _id: "$employee_id",
    projects: { $addToSet: "$project_id" }
  },
},
{
}
$match: {
  projects: { $all: rndProjects }
}
])
```

OUTPUT:

```
[
]
```

(No employee works on all R&D projects)

12.

```
db.Assignments.aggregate([
{
  $group: {
```

```

_id: "$employee_id",
projects: { $addToSet: "$project_id" }
}
},
{
$project: {
projectCount: { $size: "$projects" }
}
},
{
}
])
$match: {
projectCount: { $gt: 1 }
}

```

OUTPUT:

```

[
Employees belong mainly to:
Engineering,
R&D
]

```

13.

```

db.Assignments.aggregate([
{
$match: {
hours_worked: { $lt: 10 }
}
}
]
)

```

```
}  
  
,  
  
{  
  $lookup: {  
    from: "Employees",  
    localField: "employee_id",  
    foreignField: "employee_id",  
    as: "employee"  
  }  
  
,  
  
{  
  $lookup: {  
    from: "Projects",  
    localField: "project_id",  
    foreignField: "project_id",  
    as: "project"  
  }  
  
,  
  
{ $unwind: "$employee" },  
{ $unwind: "$project" }  
])
```

OUTPUT:

```
[  
  
{  
  
}  
  
]
```



```
employee_name: "George King",  
project_name: "Q4 Sales Strategy"
```

14.

```
db.Projects.aggregate([  
  {  
    $lookup: {  
      from: "Assignments",  
      localField: "project_id",  
      foreignField: "project_id",  
      as: "assignments"  
    }  
  },  
  {  
  }  
])  
$match: {  
  assignments: { $size: 0 }  
}
```

OUTPUT:

```
[  
  { project_name: "Internal HR Tool" }  
]  
*
```

15.

```
db.Employees.aggregate([  
  {
```

```
}  
])  
$group: {  
  _id: "$dept_id",  
  maxSalary: { $max: "$salary" }  
}
```

OUTPUT:

```
[  
]
```

(No employee exceeds department maximum salary)

16.

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$employee_id",  
      oldest_project: { $min: "$start_date" }  
    }  
  }  
])
```

OUTPUT:

```
[  
  { employee_id: 109, oldest_project: ISODate("2022-11-01") },  
  { employee_id: 101, oldest_project: ISODate("2022-11-01") }  
]
```

17.

```
db.Projects.find({
```

```
start_date: {
  $gte: ISODate("2023-01-01"),
  $lt: ISODate("2024-01-01")
}
})
```

OUTPUT:

```
[
  { project_name: "Aurora Migration" },
  { project_name: "Synergy App Dev" }
]
```

18.

```
db.Employees.aggregate([
  {
    $project: {
      _id: 0,
      employee_name: 1,
      manager_status: {
        $ifNull: ["$manager_id", "CEO"]
      }
    }
  }
])
```

OUTPUT:

```
[
  { employee_name: "Alice Smith (CEO)", manager_status: "CEO" },
  { employee_name: "Bob Johnson", manager_status: 100 }
]
```

```
]
```

19.

```
db.Assignments.aggregate([
{
  $lookup: {
    from: "Employees",
    localField: "employee_id",
    foreignField: "employee_id",
    as: "employee"
  },
  { $unwind: "$employee" },
  {
    $group: {
      _id: "$employee.dept_id",
      total_hours: { $sum: "$hours_worked" }
    },
    {
      $sort: {
        total_hours: -1
      }
    },
  }
])
```

\$limit: 3

OUTPUT:

```
[  
  { department: "Engineering", total_hours: 1080 },  
  { department: "R&D", total_hours: 235 },  
  { department: "Consulting", total_hours: 20 }  
]
```

20.

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$project_id",  
      total_hours: { $sum: "$hours_worked" }  
    }  
  },  
  {  
    $project: {  
      status: {  
        $cond: {  
          if: { $gt: ["$total_hours", 300] },  
          then: "High Activity",  
          else: "Normal"  
        }  
      }  
    }  
  }  
])
```

```
)
```

OUTPUT:

```
[  
  { project_id: 5001, status: "High Activity" },  
  { project_id: 5002, status: "Normal" },  
  { project_id: 5006, status: "High Activity" }  
]
```

21.

```
db.Employees.aggregate([  
  {  
    $lookup: {  
      from: "Employees",  
      localField: "manager_id",  
      foreignField: "employee_id",  
      as: "manager"  
    }  
  },  
  {  
    $project: {  
      _id: 0,  
      employee_name: 1,  
      manager_name: {  
        $ifNull: [  
          { $arrayElemAt: ["$manager.employee_name", 0] },  
          "No Manager"  
        ]  
      }  
    }  
  }  
])
```

```
}  
}  
}  
])
```

OUTPUT:

```
[  
  { employee_name: "Alice Smith (CEO)", manager_name: "No Manager" },  
  { employee_name: "Bob Johnson", manager_name: "Alice Smith (CEO)" }  
]
```

22.

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$project_id",  
      uniqueEmployees: { $addToSet: "$employee_id" }  
    }  
  },  
  {  
  }  
  $project: {  
    employee_count: { $size: "$uniqueEmployees" }  
  }  
])
```

OUTPUT:

```
[  
  { project_id: 5001, employee_count: 3 },
```

```
{ project_id: 5002, employee_count: 3 },  
{ project_id: 5006, employee_count: 3 }  
]
```

23.

```
var projects101 = db.Assignments.find(  
  { employee_id: 101 },  
  { project_id: 1, _id: 0 }  
)  
.toArray().map(p => p.project_id);  
db.Assignments.aggregate([  
  {  
  }  
])  
$match: {  
  project_id: { $in: projects101 },  
  employee_id: { $ne: 101 }  
}
```

OUTPUT:

```
[  
  "Charlie Brown",  
  "Fiona Glen",  
  "Jane Doe"  
]
```